GG-B-120a
SEPTEMBER 20, 1956
SUPERSEDING
Int. Fed. Spec. GG-B-00120(GSA-FSS)
February 2, 1956, and
Fed. Spec. GG-B-120
November 23, 1944

FEDERAL SPECIFICATION

BASEPLATE, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers baseplates used for making rigid trial plates and individual impression trays in dental work.

1.2 Classification.

1.2.1 *Type, grade, and classes.*—Baseplates shall be of but one type, one grade, and of the following two classes, as specified in the invitation for bids:

Class A.—Upper form.
Class B.—Lower form.

2. APPLICABLE SPECIFICATIONS AND STANDARD

2.1 The following specifications and standard of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-585—Boxes; Wood, Wirebound.
PPP-B-601—Boxes; Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-676 — Boxes; Set-Up, Paperboard.

Military Specifications:

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL-L-10547 — Liners, Case, Waterproof.

Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—The baseplates shall be uniform in composition, consisting essentially

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of a combination of high-grade shellac and mica.

3.2 Adaptability.—Baseplate shall be capable of easy adaptation. It shall not tear, blister, discolor, or stick to a talc dusted test model when heated, nor warp or lift away at any point from a test model when tested in accordance with 4.3.2. It shall not char or burn when luted to a test model with a hot spatula.

3.3 Cooling.—Baseplate, after flaming or heating in oven, shall readily revert to the original hardness upon cooling.

3.4 Transverse strength.—The transverse strength shall be not less than 1.75 kg., when tested in accordance with 4.3.3.

3.5 Bending.—The specimen shall bend 10 mm., plus or minus 5 mm., when tested in accordance with 4.3.4.

3.6 Color. — Unless otherwise specified, baseplates shall be either brown or pink.

3.7 Dimensions. — Baseplates shall be of standard commercial shapes and sizes and shall have the following dimensions (see table I).

TABLE I.

Class	Thickness	Diameter across heel	Diameter front to back
	mm.	*mm.*	*mm.*
A	1.27	98	69
B	1.27	98	71

3.7.1 *Tolerances.*—An allowance of plus or minus 0.1 mm. will be allowed on thickness and plus or minus 1.0 mm. on other dimensions.

3.8 Workmanship.—The baseplates shall be free from blisters, cracks or crinkles. Both surfaces of the baseplates shall be smooth.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Not less than one package of each class of baseplate material delivered shall be taken at random from each delivery for inspection and tests.

4.2 Inspection.—Samples shall be inspected to determine compliance with specification requirements.

4.3 Tests.

4.3.1 *Test specimens.* — Cut 1-inch-wide strips from the longest section of an upper baseplate by scoring baseplate deeply with a sharp razor blade. Break along the scored lines and sand the edges smooth by rubbing strip back and forth over fine emery paper on a flat surface.

4.3.2 *Adaptability.* — A test specimen of baseplate material shall be placed on a test model that has been preheated (to 90°C. plus or minus 2°C. for at least 1 hour) and placed in an oven for 10 minutes at 90°C., plus or minus 2°C. Remove test model and specimen and allow to cool to room temperature for 1 hour. The test model shall be constructed of maple wood to the dimensions as given in figure 1.

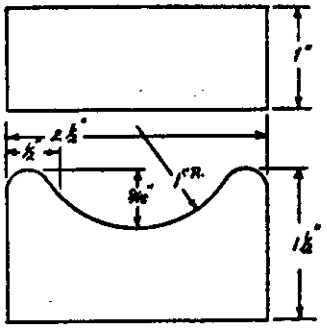


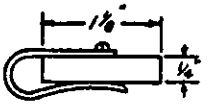
FIGURE 1.

4.3.3 *Transverse strength.*—A test specimen of baseplate material shall be suspended across a 2-inch bridge using $\frac{3}{16}$-inch glass rods for supports. Weight shall be applied in the middle of the strip by a plunger with its end consisting of an edge $\frac{3}{32}$-inch radius

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and ½ inch long. Shot shall be poured into a can on top of the plunger at a rate of 1.5 kg. per minute until the specimen breaks. An average of three weights of the plunger and shot shall be taken as the transverse strength.

4.3.4 *Bending.*—Place a whole upper plate on a smooth flat surface and flame with a Bunsen burner until softened, turn over and flame again until softened. Put aside to set at room temperature (21° to 27°C.) for exactly 1 hour. After 1 hour, place the baseplate in an oven at 37°C. plus or minus ½°C. for exactly 10 minutes. The baseplate shall stand on edge in a narrow slot so that it will be heated evenly. When heated for 10 minutes place the plate in a horizontal position with one slanting edge inserted into a slot 7/16 inch deep and just a little thicker than a baseplate to permit easy insertion and removal. Attach a weight and spring (fig. 2) totaling 34 to 35 grams to the baseplate at the furthest point from the slot so that the edge of the weight enclosed by the spring shall be flush with the edge of the baseplate. The slot shall be above the floor of the oven a sufficient distance to allow for sagging of the baseplate when heated. The oven shall be regulated so that the temperature is constant at 37°C. when thermometer bulb is at the same level as the baseplate. The baseplate shall remain under the weight for exactly 5 minutes and shall then be removed to room temperature to cool until rigid. Measure with a scale the number of millimeters the edge of the plate has warped from the horizontal. This shall be done by placing the flat surface which was in the slot on a glass plate so that the warped section extends upward. Measure the height of the other edge from the glass.



NOTE:
WIDTH OF CLIP 1/8"
MATERIAL - BRASS

TEST BLOCKS
Dental baseplate

FIGURE 2.

5. PREPARATION FOR DELIVERY

(For definitions and levels, see 6.1.)

5.1 Packaging.

5.1.1 *Unit of issue.*—Twelve (12) baseplates, of the class specified, constitute one unit.

5.1.2 *Quantities.*—Packaging and packing quantities shall be as specified in table II.

TABLE II.—*Packaging and packing quantities*

Unit pack	Intermediate pack	Shipping container
1 unit	12 units	144 units

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit carton.*—Twelve (12) baseplates (1 unit) shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–676. Carton shall contain sufficient cushioning material to prevent shifting and breakage. Carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.*—The number of units specified in table II shall be packaged in a carton of appropriate size and design, or shall be completely wrapped with

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Kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Each unit shall be packaged in standard commercial containers of the size and kind commonly used for the purpose which will afford the degree of protection required for shipment and use of the end product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.* — The number of units specified in table II shall be overpacked in an exterior container designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; Federal Specification PPP–B–601, table II, using type III, class 1 plywood; Federal Specification PPP–B–621; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping (when required) shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—The number of units specified in table II shall be overpacked in an exterior container conforming to Federal Specification PPP–B–601, table I, using class 1 plywood; or an exterior container designed for type 2 load and constructed in accordance with Federal Specification PPP–B–621; PPP–B–585, style 3, for class 2 use; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.*—The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Unit cartons.*—Each unit carton shall be marked with stock number, item identification, unit, quantity, lot (control) number and name of contractor.

5.3.2 *Intermediate packages and shipping container.*—Each intermediate package and each shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color marking and symbol are not required on containers for level A packs when addressed to domestic establishments for the Medical Services of the Armed Forces.

6. NOTES

6.1 Federal Standard 102 shall be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Purchasers should exercise any desired options offered herein and specify class required. (See 1.2.1, 3.6, 5.1 and 5.2).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular pur-

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pose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the title of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 **Transportation description.**—Transportation description applicable to this item is:

Dental goods, not otherwise indexed by name.

Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Truckload minimum weight 12,000 pounds.

6.6 This specification covers the following items appearing in the Armed Services Medical Stock List:

6520–501–6050—Baseplate, Dental, Lower Form, 12s.

6520–501–6150—Baseplate, Dental, Upper Form, 12s.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 10 cents each.

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